

AI for Research & Science

Three use-cases



Presentation at the NEO-AMELEAS Workshop

Louis HAUSEUX (Inria Sophia Antipolis, 3IA-DS4H scholarship from Université Côte d'Azur)

Supervised by Konstantin AVRACHENKOV (NEO) & Josiane ZERUBIA (AYANA)



Outline

I) AI for Skeptical People

- Creating Vector Images with TikZ
- Saving Time with AI: Creating a Counterexample

II) AI for Enthusiasts

- GitHub, Codespaces, and Antigravity (Gemini-CLI, Google)

01

AI for Skeptical People



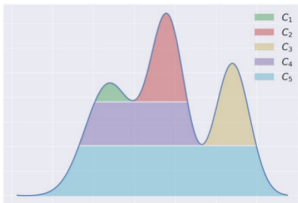
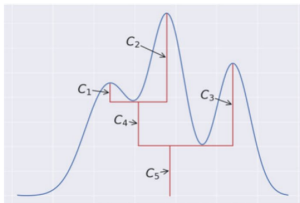


Enhancing Visual Quality: Fast Vector Graphics with AI

- ▶ **Fast & Professional Figures:** AI allows you to generate high-quality illustrations quickly without wasting time.
- ▶ **What is TikZ?:** A native LaTeX programming language for vector graphics that integrates perfectly with your document.
- ▶ **Full Control:** Since TikZ figures are code-based, you always keep full control to manually edit or fine-tune any detail at any time.



Example: Target Raster Image

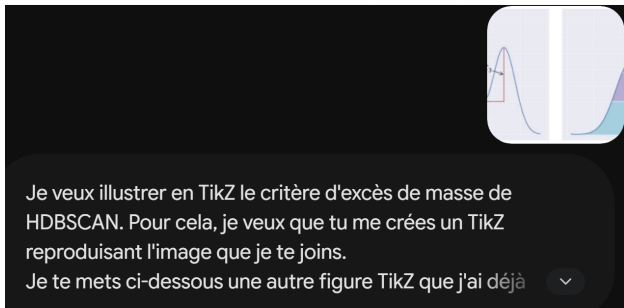


The Challenge:

- ▶ A 1D density plot illustrating the *excess of mass* in HDBSCAN.
- ▶ Simple raster formats (PNG/JPG) are pixelated and cannot match document fonts.



Example: Prompting the AI



Human-AI Collaboration:

- ▶ Simply upload the target image and ask the AI to reproduce it in TikZ.
- ▶ Saves hours of calculating coordinates and writing syntax manually.



Example: The Generated TikZ Code

TikZ Code Structure:

```
\begin{tikzpicture}[scale=0.45]
  \tikzset{declare function={f(x) = ...}}

  % Draw density curve
  \draw[thick, blue] plot[domain=0:8] ...;

  % Draw cluster tree bars
  \draw[very thick, red] (1.87,1.70) -- ...;

  % Add labels
  \draw[<-, semithick] (1.87,1.82) -- ...;
\end{tikzpicture}
```

LaTeX Integration:

► Include Package:

- Add `\usepackage{tikz}` in your preamble.

► Direct Placement:

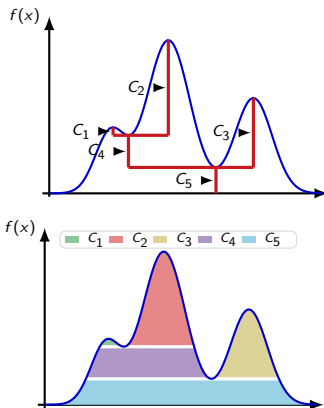
- Paste the generated code directly in the document body.

► No Assets to Manage:

- No need for external image files (PNG/PDF).



Example: The Native TikZ Result



Why it works:

- ▶ Native vector rendering means infinite scaling and crispness.
- ▶ Complete control over font size and styling.



Saving Time with AI: Creating a Counterexample



02

AI for Enthusiasts





GitHub, Codespaces, and Antigravity (Gemini-CLI, Google)



Thank you!